

COSC2129/2723/3118
Artificial Intelligence

**Reasoning under Uncertainty:
Bayesian Networks**

Road Map for Today

- ❑ Recap of probability theory
- ❑ Independence and conditional independence
- ❑ Bayes' rule
- ❑ Bayesian networks
 - Syntax and semantics
 - Compactness and node ordering
 - Construction of Bayesian networks

Why Bother Probability Theory?

- ❑ Logical reasoning may break down when dealing with uncertainty, e.g., dental diagnosis; It fails mainly because of **laziness** and **ignorance**.
- ❑ Probability theory deals with **degrees of belief** instead of degrees of truth; Probability summarizes uncertainty coming from laziness and ignorance.
- ❑ Utility theory (representing and reasoning with preferences): every state has a degree of usefulness (i.e., utility) to an agent, and the agent will prefer states with higher utility.
- ❑ Decision Theory = Probability Theory + Utility Theory
- ❑ Principle of maximum expected utility (MEU): an agent is rational if and only if it choose the action that yields the highest expected utility, averaged over all the possible outcomes of the action.

Basic Concepts in Probability Theory

- ❑ Sample space and events
- ❑ Random variables and their domains
- ❑ Probability vs. probability distribution
 - For a Boolean variable (or proposition) A , $A = \text{true}$ ($A = \text{false}$) can be abbreviated as a ($\neg a$), and thus $P(A = \text{true})$ can be represented as $P(a)$.
 - $P(\text{Weather} = \text{sunny}) = 0.72$ can be abbreviated as $P(\text{sunny}) = 0.72$.
 - $P(\text{Weather}) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$ (normalized, i.e., sums to 1)
- ❑ Prior/unconditional probability (distribution) vs. posterior/conditional probability (distribution)
- ❑ Joint probability (distribution)
- ❑ Conditional probability definition and its equivalent **product rule**
- ❑ Conditioning $\neq$ Logical Implication

Probability Axioms

$0 \leq P(\omega) \leq 1$, and $\sum_{\omega \in \Omega} P(\omega) = 1$ where Ω is the sample space

The probability of an event A (a subset of Ω): $P(A) = \sum_{\omega \in A} P(\omega)$

$$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$$

$$P(\text{true}) = 1$$

$$P(\text{false}) = 0$$

Inference by Enumeration

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

For any proposition ϕ , summing over all possible worlds where it holds:

$$P(\phi) = \sum_{\omega \in \mathcal{M}(\phi)} P(\omega)$$

$$P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$

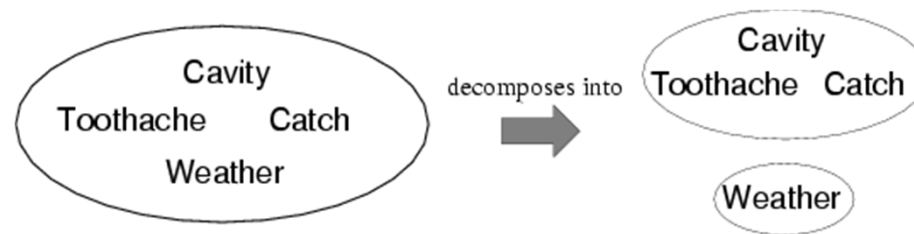
$$P(\text{cavity} \vee \text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 + 0.072 + 0.008 = 0.28$$

$$\begin{aligned} P(\neg \text{cavity} \mid \text{toothache}) &= \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})} \\ &= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4 \end{aligned}$$

Independence

Variables X and Y are independent if and only if

$$P(A | B) = P(A) \quad \text{or} \quad P(B | A) = P(B) \quad \text{or} \quad P(A, B) = P(A) P(B)$$



Absolution
(or marginal)
Independence

$$\begin{aligned} P(\textit{Toothache}, \textit{Catch}, \textit{Cavity}, \textit{Weather}) \\ &= P(\textit{Weather} | \textit{Toothache}, \textit{Catch}, \textit{Cavity}) P(\textit{Toothache}, \textit{Catch}, \textit{Cavity}) \\ &= P(\textit{Weather}) P(\textit{Toothache}, \textit{Catch}, \textit{Cavity}) \end{aligned}$$

Independence assertions are usually based on domain knowledge; When available, they can reduce the size of the domain representation.

32 entries reduce to 12 ones (4 + 8); for n independent coin flips: $O(2^n) \rightarrow O(n)$

Absolute independence is powerful but quite rare.

Independent subsets can be very large, e.g., dentistry.

Conditional Independence

$P(\textit{Toothache}, \textit{Cavity}, \textit{Catch})$ has $2^3 - 1 = 7$ independent entries

If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:

$$(1) P(\textit{catch} \mid \textit{toothache}, \textit{cavity}) = P(\textit{catch} \mid \textit{cavity})$$

The same independence holds if I do not have a cavity:

$$(2) P(\textit{catch} \mid \textit{toothache}, \neg \textit{cavity}) = P(\textit{catch} \mid \neg \textit{cavity})$$

Catch is conditionally independent of *Toothache* given *Cavity*.

$$P(\textit{Catch} \mid \textit{Toothache}, \textit{Cavity}) = P(\textit{Catch} \mid \textit{Cavity})$$

Equivalent statements:

$$P(\textit{Toothache} \mid \textit{Catch}, \textit{Cavity}) = P(\textit{Toothache} \mid \textit{Cavity})$$

$$P(\textit{Toothache}, \textit{Catch} \mid \textit{Cavity}) = P(\textit{Toothache} \mid \textit{Cavity}) P(\textit{Catch} \mid \textit{Cavity})$$

Conditional Independence (cont'd)

Write out the full joint distribution using chain rule:

$$\begin{aligned} & P(\textit{Toothache}, \textit{Catch}, \textit{Cavity}) \\ &= P(\textit{Toothache} \mid \textit{Catch}, \textit{Cavity}) P(\textit{Catch}, \textit{Cavity}) \\ &= P(\textit{Toothache} \mid \textit{Catch}, \textit{Cavity}) P(\textit{Catch} \mid \textit{Cavity}) P(\textit{Cavity}) \\ &= P(\textit{Toothache} \mid \textit{Cavity}) P(\textit{Catch} \mid \textit{Cavity}) P(\textit{Cavity}) \end{aligned}$$

i.e., $2 + 2 + 1 = 5$ independent numbers

Conditional independence assertions can allow probabilistic systems to scale up.

Conditional independence assertions are much more commonly available than absolute independence assertions, e.g., *Cavity* separates *Toothache* and *Catch* because it is a direct cause of both of them.

Bayes' Rule

Product rule: $P(a \wedge b) = P(a | b) P(b) = P(b | a) P(a)$

$$\Rightarrow \text{Bayes' rule: } P(b | a) = \frac{P(a | b) P(b)}{P(a)}$$

or in distribution form

$$P(Y | X) = \frac{P(X | Y) P(Y)}{P(X)} = \alpha P(X | Y) P(Y)$$

Useful for assessing *diagnostic* probability from *causal* probability:

$$P(\text{Cause} | \text{Effect}) = \frac{P(\text{Effect} | \text{Cause}) P(\text{Cause})}{P(\text{Effect})}$$

E.g., let M be meningitis, S be stiff neck:

$$P(m | s) = P(s | m) P(m) / P(s) = 0.8 \times 0.0001 / 0.1 = 0.0008$$

Note: the posterior probability of meningitis is very small!



Bayes' Rule and Conditional Independence

$$\begin{aligned} P(\text{Cavity} \mid \text{toothache} \wedge \text{catch}) \\ &= \alpha P(\text{toothache} \wedge \text{catch} \mid \text{Cavity}) P(\text{Cavity}) \\ &= \alpha P(\text{toothache} \mid \text{Cavity}) P(\text{catch} \mid \text{Cavity}) P(\text{Cavity}) \end{aligned}$$

This is an example of a naïve Bayes model as follows:

$$P(\text{Cause}, \text{Effect}_1, \dots, \text{Effect}_n) = P(\text{Cause}) \prod_i P(\text{Effect}_i \mid \text{Cause})$$



Total number of parameters is *linear* in n

Bayesian Networks

- ❑ The full joint probability distribution can answer any queries but become too large as the number of variables grows.
- ❑ Specifying probabilities for each possible world is practically impossible.
- ❑ (Conditional) independence relationships among variables can greatly reduce the number of probabilities to be specified to define the full joint distribution.
- ❑ **Bayesian network** is a graph data structure representing dependencies among variable, which can implicitly represent any full joint distributions concisely.
- ❑ Syntax of Bayesian networks:
 - a set of nodes, one per variable
 - a **directed acyclic graph** (directed links $\approx$ "directly influences")
 - a conditional distribution for each node given its parents:

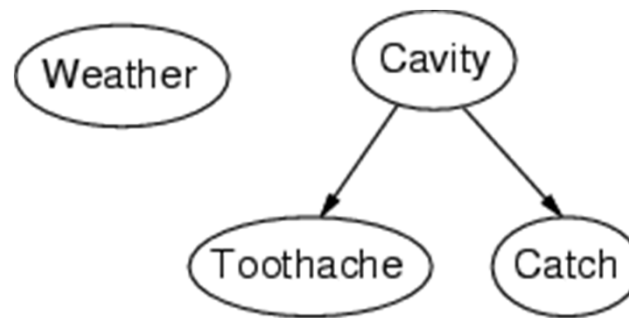
$$P(X_i | \text{Parents}(X_i))$$

Bayesian Networks (cont'd)

- ❑ For discrete variables, conditional distributions are usually represented as **conditional probability tables** (CPTs):
 - Each row contains the probabilities of each node value for a **conditioning case**, summing to 1.
 - Conditioning case: a possible combination of values for the parents nodes.
 - Boolean variable has only one element per row.
 - Nodes with no parents only have one row (prior probabilities).
- ❑ Network topology specifies the conditional independence relationships that hold in the domain.
- ❑ **Topology + conditional distributions** suffices to specify the full joint distribution for all the variable.

Dentistry Example

The topology of network encodes conditional independence assertions:



Weather is independent of the other variables

Toothache and *Catch* are conditionally independent, given *Cavity*

Burglary Example

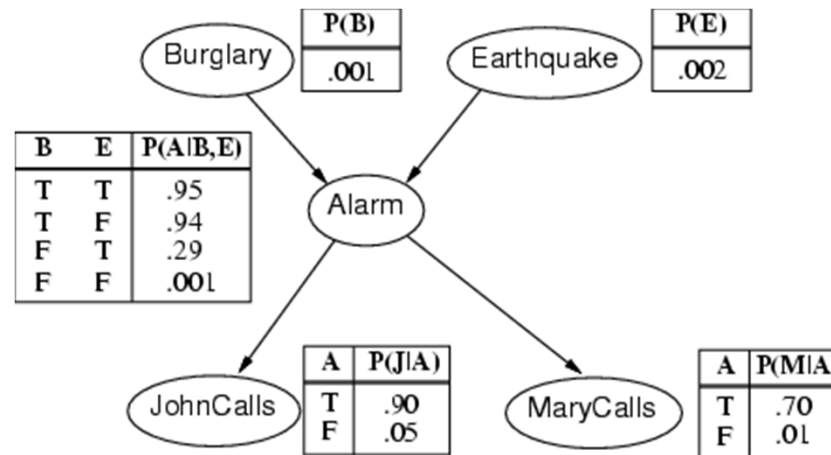
I have installed a new burglar alarm which can reliably detect a burglary and also responds to minor earthquakes. I have two neighbors: John and Mary. They have promised to call me at work when they hear the alarm. John nearly always call one he hears the alarm but he often confuses the telephone ring with the alarm. Mary often misses the alarm because she likes listening very loud music.

I'm at work, neighbor John calls to say my alarm is on, but neighbor Mary doesn't call. What is the probability of a burglary?



Variables: *Burglary, Earthquake, Alarm, JohnCalls, MaryCalls*

Burglary Example (cont'd)



“Causal” knowledge encoded in the network topology:

- A burglar can set the alarm off.
- An earthquake can set the alarm off.
- The alarm can cause Mary to call.
- The alarm can cause John to call.

Notes: Assuming that John and Mary don’t perceive burglary directly; they do not feel minor earthquakes; they do not confer before calling.

Semantics of Bayesian Networks

Semantics – what a network means:

- A (more compact) representation of the joint probability distribution; helpful in understanding how to construct network.
- Encoding a collection of conditional independence statements; helpful in understanding how to design inference procedures.

A generic entry in the joint distribution is the probability of a conjunction of particular assignments to each variable, e.g., $P(x_1, \dots, x_n)$, defined by:

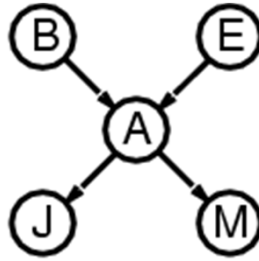
$$P(x_1, \dots, x_n) = \prod_{i=1}^n \theta(x_i \mid \text{parents}(X_i))$$

where $\text{parents}(X_i)$ denotes the values of $\text{Parents}(X_i)$ that appear in $x_1, \dots, x_n$.

From the above definition, it can be proved that:

$$\theta(x_i \mid \text{parents}(X_i)) = P(x_i \mid \text{parents}(X_i))$$

Compactness



B - Burglary;
E - Earthquake;
A - Alarm;
J - JohnCalls;
M - MaryCalls;

A CPT for Boolean variable X_i with k Boolean parents has 2^k rows for all possible combinations of parent values.

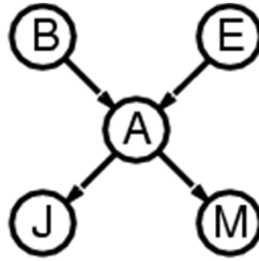
Each row requires one number p for $P(X_i = \text{true})$
(the number for $P(X_i = \text{false})$ is just $1-p$)

If each variable has no more than k parents, the complete network requires $O(n \cdot 2^k)$ numbers.

i.e., grows linearly with n , vs. $O(2^n)$ for the full joint distribution

For burglary example, $1 + 1 + 4 + 2 + 2 = 10$ numbers (vs. $2^5 - 1 = 31$).

Global Semantics



B - Burglary;
E - Earthquake;
A - Alarm;
J - JohnCalls;
M - MaryCalls;

The full joint distribution is defined as the product of local conditional distributions:

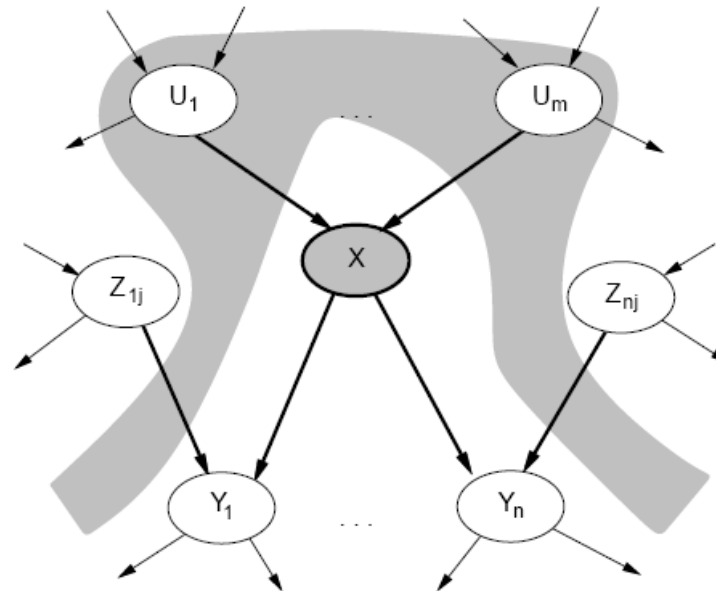
$$\begin{aligned} P(x_1, x_2, \dots, x_n) &= P(x_1) \times P(x_2 \mid x_1) \times \dots \times P(x_n \mid x_1 \wedge \dots \wedge x_{n-1}) \\ &= \prod_{i=1}^n P(x_i \mid x_1 \wedge \dots \wedge x_{i-1}) \\ &= \prod_{i=1}^n P(x_i \mid \text{parent}(X_i)) \end{aligned}$$

$$\begin{aligned} \text{e.g., } P(j \wedge m \wedge a \wedge \neg b \wedge \neg e) &= P(j \mid a) P(m \mid a) P(a \mid \neg b, \neg e) P(\neg b) P(\neg e) \\ &= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998 \\ &= 0.00062 \end{aligned}$$

Local Semantics

Local semantics: each node is conditionally independent of its non-descendants, given its parents.

E.g., *JohnCalls* is independent of *Burglary* and *Earthquake*, given the value of *Alarm*.

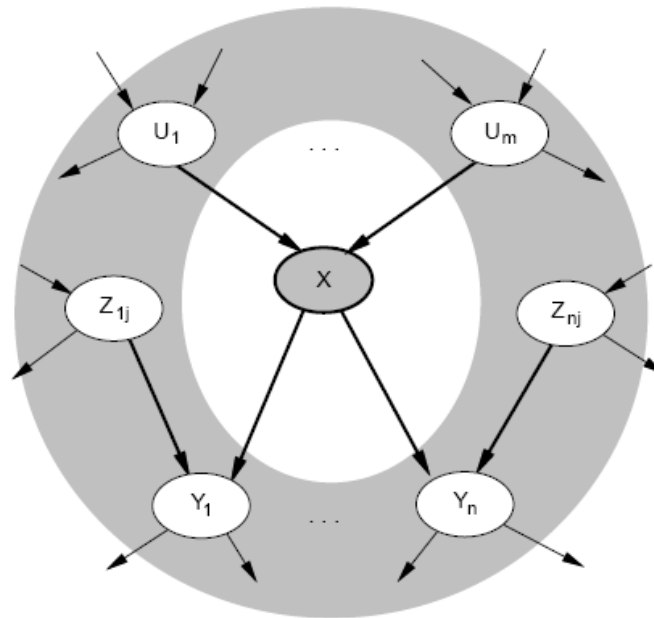


Theorem: Local semantics $\Leftrightarrow$ global semantics

Markov Blanket

Each node is conditionally independent of all others given its Markov blanket:

parents + children + children's parents



Constructing Bayesian Networks

$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid \text{parents}(X_i))$ defines what a given Bayesian network means.

How to construct a Bayesian network so that the resulting joint distribution is a good representation of a given domain?

According to the chain rule:

$$\begin{aligned} P(x_1, \dots, x_n) &= P(x_n \mid x_{n-1}, \dots, x_1) P(x_{n-1} \mid x_{n-2}, \dots, x_1) \dots P(x_2 \mid x_1) P(x_1) \\ &= \prod_{i=1}^n P(x_i \mid x_{i-1}, \dots, x_1) \end{aligned}$$

We have $\mathbf{P}(X_i \mid X_{i-1}, \dots, X_1) = \mathbf{P}(X_i \mid \text{Parents}(X_i))$ provided that $\text{Parents}(X_i) \subseteq \{X_{i-1}, \dots, X_1\}$.

Bayesian network is a correct representation only if each node is conditionally independent of its other predecessors in the node ordering, given its parents.

Constructing Bayesian Networks

Need a method such that a set of conditional independence relationships guarantees the required global semantics:

1. Choose a set of relevant variables X_i that describe the domain.
2. Choose an ordering for the variables. (**causes precedes effects**).
3. For $i = 1$ to n *do*:
 - a) Pick variable X_i and add a node to the network for it.
 - b) Set $\text{Parent}(X_i)$ to a minimal set chosen from $\{X_1, \dots, X_{i-1}\}$ such that the conditional independence property is satisfied:

$$\mathbf{P}(X_i \mid X_1, \dots, X_{i-1}) = \mathbf{P}(X_i \mid \text{Parent}(X_i))$$

- c) Insert a directed link from each determined parent of X_i to X_i .
 - d) Write down the CPT for X_i .

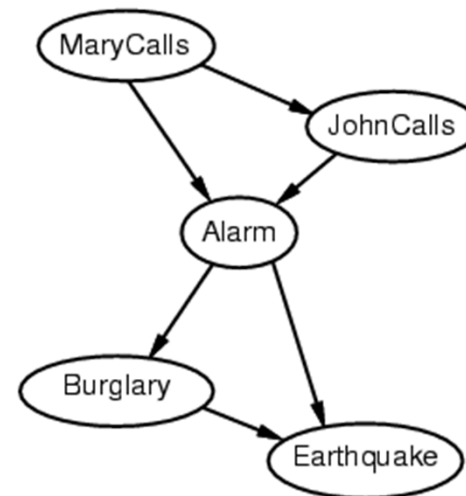
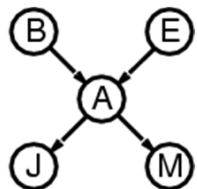
Compactness and Node Ordering

Compactness of Bayesian Networks is an example of a general property of locally structured (or sparse) systems, which depends on the proper choice of the node ordering.

The correct order to add nodes is to add the “root causes” first, then the variables they influence, so on until “leaves” reached.

An example of wrong ordering:

- Suppose we choose the ordering M, J, A, B, E
- The network is less compact compared to:

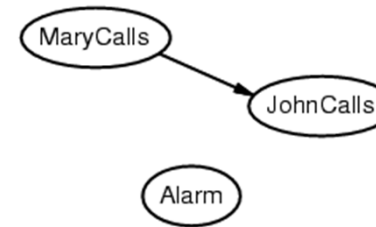


Example: Burglary Alarm

Suppose we choose the ordering M, J, A, B, E

$P(J \mid M) = P(J)$? **No**

$P(A \mid J, M) = P(A \mid J)$? $P(A \mid J, M) = P(A)$?



Example: Burglary Alarm (cont'd)

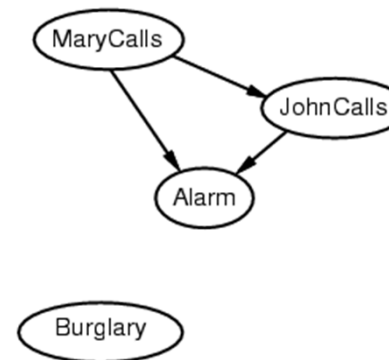
Suppose we choose the ordering M, J, A, B, E

$P(J \mid M) = P(J)$? No

$P(A \mid J, M) = P(A \mid J)$? $P(A \mid J, M) = P(A)$? No

$P(B \mid A, J, M) = P(B \mid A)$?

$P(B \mid A, J, M) = P(B)$?



Example: Burglary Alarm (cont'd)

Suppose we choose the ordering M, J, A, B, E

$P(J \mid M) = P(J)$? **No**

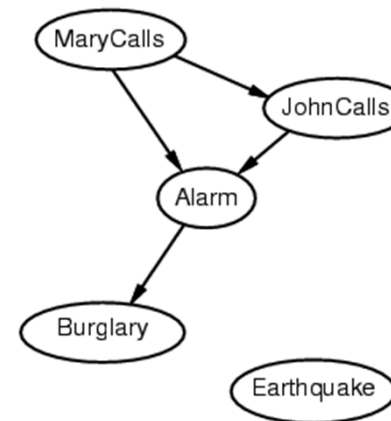
$P(A \mid J, M) = P(A \mid J)$? $P(A \mid J, M) = P(A)$? **No**

$P(B \mid A, J, M) = P(B \mid A)$? **Yes**

$P(B \mid A, J, M) = P(B)$? **No**

$P(E \mid B, A, J, M) = P(E \mid A)$?

$P(E \mid B, A, J, M) = P(E \mid A, B)$?



Example: Burglary Alarm (cont'd)

Suppose we choose the ordering M, J, A, B, E

$P(J \mid M) = P(J)$? **No**

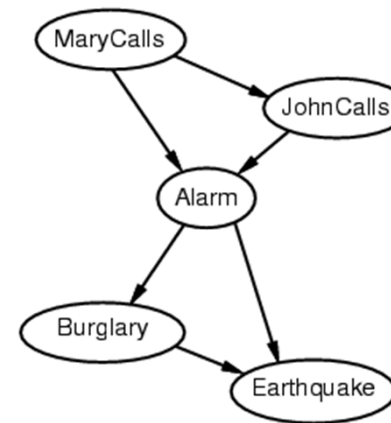
$P(A \mid J, M) = P(A \mid J)$? $P(A \mid J, M) = P(A)$? **No**

$P(B \mid A, J, M) = P(B \mid A)$? **Yes**

$P(B \mid A, J, M) = P(B)$? **No**

$P(E \mid B, A, J, M) = P(E \mid A)$? **No**

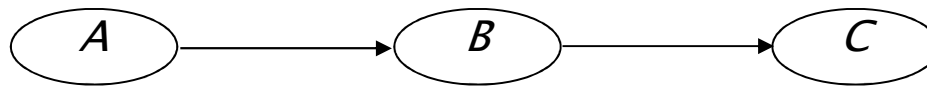
$P(E \mid B, A, J, M) = P(E \mid A, B)$? **Yes**



- Deciding conditional independence is hard in non-causal directions.
- Network is less compact: $1 + 2 + 4 + 2 + 4 = 13$ numbers are needed.

Conditional Independence: Causal Chains

Causal chains give rise to conditional independence:



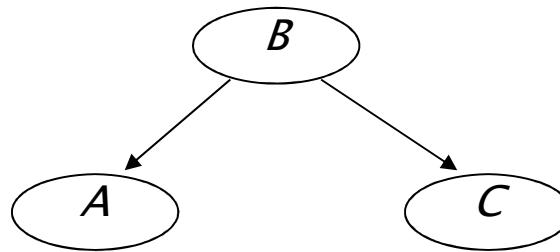
$$P(C \mid A, B) = P(C \mid B)$$

Example:

- A = excessive exercises
- B = dehydrated
- C = hard to concentrate

Conditional Independence: Common Causes

Common causes (or ancestors) give rise to conditional independence:



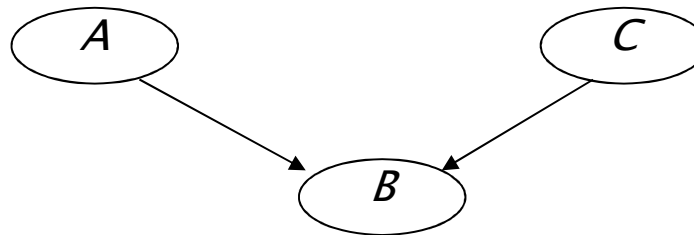
$$P(C | A, B) = P(C | B)$$

Example:

- A = thirsty
- B = dehydrated
- C = hard to concentrate

Conditional Independence: Common Effects

Common effects (or descendants) give rise to conditional dependence:



$$P(C \mid A, B) \neq P(C \mid B)$$

Example:

- A = excessive exercises
- B = dehydrated
- C = inadequate water intake

Conclusion

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- ❑ Bayes' rule
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